
Applications of Maximum and Minimum Values

Complete English Coloured L^AT_EX Edition

58 Optimization Problems

Every question translated into English and followed by a complete calculus solution

Algebraic models • Geometry • Solids • Cost and revenue • Viewing angles

A structured resource for differential-calculus applications

Clear modelling, derivative tests, exact answers, and professionally redrawn diagrams

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Optimization Method and Essential Notes

A reliable four-step method

1. Identify the quantity to be maximized or minimized and write it as the *objective function*.
2. Use the given constraints to express the objective function in one variable.
3. Differentiate, locate admissible critical points, and compare them with relevant endpoints.
4. State the requested dimensions, coordinates, time, cost, area, or volume with units.

Language clues: “largest possible,” “greatest,” and “maximum” indicate maximization; “least possible,” “minimum,” and “lowest cost” indicate minimization.

Derivative test used throughout. If the derivative changes from positive to negative at a critical point, the objective has a local maximum there. If it changes from negative to positive, it has a local minimum. On a closed physical interval, endpoints must also be checked when applicable.

Rectangles, Curves, Motion, and Coordinate Geometry

Problem 1

A rectangular plot of land has perimeter 600 m. Find its dimensions if its area is to be as large as possible.

Solution 1

Let the side lengths be x and y . Then $2x + 2y = 600$, so $y = 300 - x$.

$$A(x) = x(300 - x) = 300x - x^2, \quad 0 < x < 300.$$

Differentiate and test. $A'(x) = 300 - 2x$, so $A'(x) = 0$ at $x = 150$. Since $A''(x) = -2 < 0$, this gives the maximum.

Result: The plot must be a square: 150 m $\times$ 150 m, with maximum area 22,500 m².

Problem 2

A rectangular plot has area 400 m². Find the smallest possible perimeter.

Solution 2

Let the sides be x and y . Since $xy = 400$, we have $y = 400/x$.

$$P(x) = 2x + 2y = 2x + \frac{800}{x}, \quad x > 0.$$

Differentiate and test. $P'(x) = 2 - 800/x^2$. Hence $P'(x) = 0$ when $x^2 = 400$, so $x = 20$. Then $y = 20$. Also $P''(x) = 1600/x^3 > 0$, so this is a minimum.

Result: The least perimeter is 80 m, attained by a 20 m $\times$ 20 m square.

Problem 3

Two positive numbers have sum 20.

- (a) Find the numbers for which the sum of their squares is as small as possible.
 (b) Find the numbers for which the product of one number and the square of the other is as large as possible.

Solution 3

Let the numbers be x and $20 - x$, where $0 < x < 20$.

(a)

$$S(x) = x^2 + (20 - x)^2 = 2x^2 - 40x + 400.$$

Then $S'(x) = 4x - 40 = 0$ gives $x = 10$, and $S''(x) = 4 > 0$.

(b) Take the squared number to be y and the other number to be $20 - y$:

$$P(y) = (20 - y)y^2 = 20y^2 - y^3.$$

Then $P'(y) = 40y - 3y^2 = y(40 - 3y)$, so the interior critical point is $y = 40/3$. The other number is $20/3$, and $P''(40/3) < 0$.

Result: (a) 10 and 10. (b) The squared number is $40/3$ and the other is $20/3$; the maximum product is $16,000/27$. The roles may be interchanged.

Problem 4

Find the x -coordinate of the point at which the tangent slope to

$$y = \frac{1}{3}x^3 - 3x^2 - 40$$

is as small as possible.

Solution 4

The slope function is

$$m(x) = y'(x) = x^2 - 6x = (x - 3)^2 - 9.$$

This quadratic has its minimum at $x = 3$.

Result: The required x -coordinate is $\boxed{3}$. The minimum tangent slope is -9 .

Problem 5

A particle moves along a straight line according to

$$s(t) = t^4 - 12t^3 + 78t^2 - 6t + 30,$$

where s is measured in metres and t in seconds. Find the least acceleration of the particle.

Solution 5

Velocity and acceleration are

$$v(t) = s'(t) = 4t^3 - 36t^2 + 156t - 6,$$

$$a(t) = v'(t) = 12t^2 - 72t + 156 = 12(t - 3)^2 + 48.$$

The square term is nonnegative, so the acceleration is minimized at $t = 3$.

Result: The least acceleration is $\boxed{48 \text{ m/s}^2}$, occurring at $t = 3 \text{ s}$.

Problem 6

Find the maximum area of a rectangle drawn above the x -axis with one side on the x -axis and its two upper vertices on the curve

$$y = 12 - x^2.$$

Solution 6

By symmetry, place the upper vertices at $(\pm x, 12 - x^2)$, where $0 < x < \sqrt{12}$. The width is $2x$ and the height is $12 - x^2$.

$$A(x) = 2x(12 - x^2) = 24x - 2x^3.$$

Differentiate and test. $A'(x) = 24 - 6x^2 = 6(4 - x^2)$. Thus $x = 2$ gives the maximum.

Result: Width = 4, height = 8, and maximum area = $\boxed{32}$ square units.

Problem 7

Find the maximum area of a rectangle drawn above the x -axis with one side on the x -axis and its two upper vertices on

$$y = 8 - x^2 + 4x.$$

Solution 7

Complete the square:

$$y = 12 - (x - 2)^2.$$

Let u be the horizontal half-width measured from the symmetry line $x = 2$. Then the width is $2u$ and the height is $12 - u^2$.

$$A(u) = 2u(12 - u^2).$$

As in Problem 6, $A'(u) = 24 - 6u^2 = 0$ gives $u = 2$.

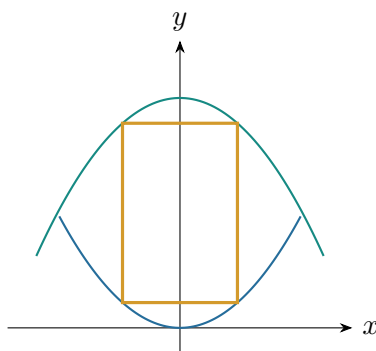
Result: The rectangle extends from $x = 0$ to $x = 4$, has height 8, and has maximum area $\boxed{32}$.

Problem 8

A rectangle is inscribed between the curves

$$y = 2x^2 \quad \text{and} \quad y = 36 - x^2,$$

with vertical sides and symmetry about the y -axis. Find the dimensions that maximize its area.



Solution 8

Let the vertical sides be at $x = \pm x$. The width is $2x$ and the height is

$$(36 - x^2) - 2x^2 = 36 - 3x^2.$$

Thus

$$A(x) = 2x(36 - 3x^2) = 72x - 6x^3.$$

Then $A'(x) = 72 - 18x^2 = 18(4 - x^2)$, so $x = 2$.

Result: The maximizing rectangle has width 4, height 24, and area 96.

Problem 9

A rectangle has its base on the x -axis. Its upper-right vertex lies in the first quadrant on

$$y = 12 - \frac{x^2}{4},$$

and its upper-left vertex lies in the second quadrant on

$$y = 12 - x^2.$$

Find the largest possible area of the rectangle.

Solution 9

Let the upper-right vertex have x -coordinate $a > 0$, and let the upper-left vertex have coordinate $-b$, where $b > 0$. Equal heights give

$$12 - \frac{a^2}{4} = 12 - b^2 \implies b = \frac{a}{2}.$$

Hence width $= a + b = 3a/2$ and height $= 12 - a^2/4$.

$$A(a) = \frac{3a}{2} \left(12 - \frac{a^2}{4} \right) = 18a - \frac{3a^3}{8}.$$

Then $A'(a) = 18 - 9a^2/8 = 0$, so $a = 4$, $b = 2$, and the height is 8.

Result: The maximum area is 48 square units, with width 6 and height 8.

Problem 10

An isosceles triangle is drawn above the x -axis. The vertex between the equal sides is at the origin, and the other two vertices lie on

$$y = 27 - x^2.$$

Find its maximum possible area.

Solution 10

The two upper vertices are $(\pm x, 27 - x^2)$. The base is $2x$ and the perpendicular height from the origin is $27 - x^2$.

$$A(x) = \frac{1}{2}(2x)(27 - x^2) = 27x - x^3.$$

Then $A'(x) = 27 - 3x^2 = 0$ gives $x = 3$.

Result: Base = 6, height = 18, and maximum area = $\boxed{54}$.

Problem 11

A trapezoid is drawn under the parabola $y = 9 - x^2$ and above the x -axis. Its lower base is the interval between the two x -intercepts, and its upper vertices are symmetric points on the parabola. Find its maximum possible area.

Solution 11

The lower base has length 6. Let the upper vertices be $(\pm x, 9 - x^2)$, so the upper base has length $2x$ and the height is $9 - x^2$.

$$A(x) = \frac{6 + 2x}{2}(9 - x^2) = (3 + x)(9 - x^2), \quad 0 \leq x \leq 3.$$

$$A'(x) = 9 - 6x - 3x^2 = -3(x - 1)(x + 3).$$

The admissible critical point is $x = 1$, and it gives the maximum.

Result: The bases are 6 and 2, the height is 8, and the maximum area is $\boxed{32}$.

Problem 12

Find the maximum area of a rectangle that can be inscribed in a semicircle of diameter 10 cm, with one side of the rectangle on the diameter.

Solution 12

The radius is 5. Let x be half the rectangle's width and y its height. Then

$$x^2 + y^2 = 25, \quad y = \sqrt{25 - x^2}.$$

The area is $A = 2xy = 2x\sqrt{25 - x^2}$. Maximizing $A^2 = 4x^2(25 - x^2)$ gives

$$\frac{d}{dx}[x^2(25 - x^2)] = 2x(25 - 2x^2) = 0,$$

so $x = 5/\sqrt{2}$ and $y = 5/\sqrt{2}$.

Result: Dimensions: $5\sqrt{2}$ cm $\times$ $\frac{5}{\sqrt{2}}$ cm; maximum area = $\boxed{25 \text{ cm}^2}$.

Problem 13

An isosceles triangle has base 8 cm and height 10 cm. A rectangle is placed inside it with two vertices on the base and two vertices on the equal sides. Find the rectangle dimensions that maximize its area.

Solution 13

Let the rectangle height be y . By similar triangles, the available width at height y is

$$w = 8 \left(1 - \frac{y}{10}\right).$$

Thus

$$A(y) = 8y - \frac{4}{5}y^2.$$

Then $A'(y) = 8 - 8y/5 = 0$ gives $y = 5$, and $w = 4$.

Result: The maximizing rectangle is 4 cm $\times$ 5 cm, with area $\boxed{20 \text{ cm}^2}$.

Problem 14

A point $A(a, b)$ lies in the first quadrant. Find the equation of the line through A that intersects the positive coordinate axes and forms a triangle of minimum area with those axes.

Solution 14

Let the intercepts be $p > a$ and $q > b$. The line has equation $x/p + y/q = 1$, and because (a, b) lies on it,

$$\frac{a}{p} + \frac{b}{q} = 1 \implies q = \frac{bp}{p-a}.$$

The triangle area is

$$A(p) = \frac{1}{2}pq = \frac{bp^2}{2(p-a)}.$$

Differentiation gives

$$A'(p) = \frac{b}{2} \frac{p(p-2a)}{(p-a)^2}.$$

Therefore $p = 2a$, and then $q = 2b$.

Result: The required line is $\boxed{\frac{x}{2a} + \frac{y}{2b} = 1}$, and the minimum triangle area is $\boxed{2ab}$.

Problem 15

Find the equation of the line through $(2, 3)$ that cuts from the first quadrant a triangle of minimum area.

Solution 15

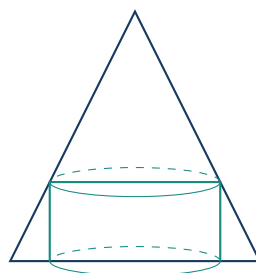
Apply Problem 14 with $a = 2$ and $b = 3$. The optimal intercepts are 4 and 6.

Result: $\boxed{\frac{x}{4} + \frac{y}{6} = 1}$, equivalently $\boxed{3x + 2y = 12}$. The minimum area is 12.

Cones, Cylinders, Spheres, and Solids of Revolution

Problem 16

A right circular cylinder is inscribed coaxially in a right circular cone of base radius 5 cm and height 9 cm. Find the cylinder dimensions that maximize its volume.



Solution 16

Let the cylinder height be h . Similar triangles give

$$\frac{r}{5} = \frac{9-h}{9}, \quad r = 5 \left(1 - \frac{h}{9}\right).$$

Hence

$$V(h) = \pi r^2 h = 25\pi h \left(1 - \frac{h}{9}\right)^2.$$

Ignoring the positive constant and differentiating,

$$\frac{d}{dh} [h(9-h)^2] = (9-h)(9-3h).$$

The interior maximum occurs at $h = 3$, and then $r = 10/3$.

Result: Radius = $\frac{10}{3}$ cm, height = 3 cm, and maximum volume = $\frac{100\pi}{3} \text{ cm}^3$.

Problem 17

Prove that the largest possible volume of a right circular cylinder inscribed coaxially in a right circular cone is $\frac{4}{9}$ of the cone's volume.

Solution 17

Let the cone have radius R and height H . If the cylinder has height h , similar triangles give

$$r = R \left(1 - \frac{h}{H}\right).$$

Thus

$$V_{\text{cyl}}(h) = \pi R^2 h \left(1 - \frac{h}{H}\right)^2.$$

As in Problem 16, the maximum occurs at $h = H/3$, so $r = 2R/3$. Therefore

$$V_{\text{max}} = \pi \left(\frac{2R}{3}\right)^2 \frac{H}{3} = \frac{4\pi R^2 H}{27}.$$

Since $V_{\text{cone}} = \pi R^2 H/3$,

$$\frac{V_{\text{max}}}{V_{\text{cone}}} = \frac{4/27}{1/3} = \frac{4}{9}.$$

Result: $V_{\text{max}} = \frac{4}{9} V_{\text{cone}}$.

Problem 18

A right circular frustum has base radii 14 cm and 5 cm and height 12 cm. A cylinder is cut from it with its lower base coinciding with the larger base and its upper circumference touching the lateral surface. Find the cylinder dimensions that maximize its volume.

Solution 18

Let h be the cylinder height. The frustum radius decreases linearly from 14 to 5, so at height h ,

$$r(h) = 14 - \frac{14-5}{12}h = 14 - \frac{3}{4}h.$$

Hence

$$V(h) = \pi h \left(14 - \frac{3}{4}h\right)^2,$$

and

$$V'(h) = \pi \left(14 - \frac{3}{4}h\right) \left(14 - \frac{9}{4}h\right).$$

The admissible interior critical point is

$$h = \frac{56}{9}, \quad r = 14 - \frac{3}{4} \cdot \frac{56}{9} = \frac{28}{3}.$$

Result: Height = $\frac{56}{9}$ cm and radius = $\frac{28}{3}$ cm. The maximum volume is $\frac{43,904\pi}{81}$ cm³.

Problem 19

A right circular cone is inscribed in a sphere of radius 9 cm. Find the cone's base radius when its volume is as large as possible.

Solution 19

Let h be the cone height measured from the sphere's top point to the base plane. In an axial cross-section,

$$r^2 = 9^2 - (9 - h)^2 = 18h - h^2.$$

Therefore

$$V(h) = \frac{\pi}{3} r^2 h = \frac{\pi}{3} h^2 (18 - h).$$

Then

$$V'(h) = \pi h(12 - h),$$

so the maximum occurs at $h = 12$. Thus

$$r^2 = 18(12) - 12^2 = 72.$$

Result: The required base radius is $6\sqrt{2}$ cm.

Problem 20

Find the height of the largest-volume right circular cylinder that can be inscribed in a sphere of radius $3\sqrt{3}$ cm.

Solution 20

Let the cylinder height be h and radius be r . From the central cross-section,

$$r^2 + \left(\frac{h}{2}\right)^2 = R^2,$$

where $R = 3\sqrt{3}$. Hence

$$V(h) = \pi r^2 h = \pi h \left(R^2 - \frac{h^2}{4}\right).$$

Then

$$V'(h) = \pi \left(R^2 - \frac{3h^2}{4}\right) = 0 \implies h = \frac{2R}{\sqrt{3}} = 6.$$

Result: The maximizing cylinder has height 6 cm and radius $3\sqrt{2}$ cm.

Problem 21

A cylindrical piece of wood has lateral surface area $400\pi \text{ cm}^2$. A hemisphere whose diameter equals the cylinder's base diameter is hollowed from it. Find the cylinder base radius that maximizes the volume of wood remaining.

Solution 21

Let the cylinder radius be r and height be h . The lateral area condition gives

$$2\pi rh = 400\pi \implies h = \frac{200}{r}.$$

The remaining volume is

$$V(r) = \pi r^2 h - \frac{2}{3}\pi r^3 = 200\pi r - \frac{2}{3}\pi r^3.$$

Then

$$V'(r) = 200\pi - 2\pi r^2 = 0 \implies r = 10.$$

Since $V''(r) = -4\pi r < 0$, this is the maximum.

Result: The optimal radius is $\boxed{10 \text{ cm}}$; then $h = 20 \text{ cm}$ and the remaining volume is $\frac{4000\pi}{3} \text{ cm}^3$.

Problem 22

A square-based rectangular prism is inscribed in a solid sphere of radius 10 cm. Find the prism dimensions that maximize its volume.

Solution 22

Let the square base side be a and the height be h . The distance from the sphere centre to a prism vertex is 10, so

$$\left(\frac{a}{2}\right)^2 + \left(\frac{a}{2}\right)^2 + \left(\frac{h}{2}\right)^2 = 100.$$

Thus

$$2a^2 + h^2 = 400, \quad a^2 = \frac{400 - h^2}{2}.$$

The volume is

$$V(h) = a^2 h = \frac{1}{2}(400h - h^3).$$

Then $V'(h) = 200 - \frac{3}{2}h^2 = 0$, giving $h = 20/\sqrt{3}$. The constraint gives $a = 20/\sqrt{3}$ as well.

Result: The maximum-volume prism is a cube of side $\boxed{\frac{20}{\sqrt{3}} \text{ cm}}$.

Problem 23

A right circular cone of base radius 4 cm and height 12 cm contains another coaxial right cone whose vertex is at the centre of the larger cone's base and whose base lies on the lateral surface of the larger cone. Find the largest possible volume of the inner cone.

Solution 23

Let the inner cone height be h , measured upward from the centre of the outer base. At height h , the outer cone's radius is

$$r = 4\left(1 - \frac{h}{12}\right) = 4 - \frac{h}{3}.$$

Thus

$$V(h) = \frac{\pi}{3}h \left(4 - \frac{h}{3}\right)^2.$$

Differentiation gives factors proportional to

$$\left(4 - \frac{h}{3}\right)(4 - h),$$

so the interior maximum occurs at $h = 4$, with $r = 8/3$.

Result: The maximum volume is $\boxed{\frac{256\pi}{27} \text{ cm}^3}$.

Problem 24

A rectangle has diagonal length 10 cm. It is revolved through one complete turn about one of its sides to form a cylinder. Find the largest possible cylinder volume.

Solution 24

Let the side used as the axis be h and the other side be r . Then

$$h^2 + r^2 = 100,$$

and the cylinder volume is

$$V = \pi r^2 h = \pi(100 - h^2)h.$$

Then

$$V'(h) = \pi(100 - 3h^2) = 0,$$

so $h = 10/\sqrt{3}$ and $r = 10\sqrt{2/3}$.

Result: $V_{\max} = \boxed{\frac{2000\pi}{3\sqrt{3}} \text{ cm}^3} = \boxed{\frac{2000\pi\sqrt{3}}{9} \text{ cm}^3}$.

Problem 25

A rectangular sheet has perimeter 36 cm. Its length is x and width is y . It is rolled so that x becomes the circumference and y becomes the height of an open cylinder. Find x and y for maximum volume.

Solution 25

The perimeter condition gives $x + y = 18$, so $y = 18 - x$. Since $x = 2\pi r$, we have $r = x/(2\pi)$. Thus

$$V(x) = \pi r^2 y = \frac{x^2(18 - x)}{4\pi}.$$

Then

$$V'(x) = \frac{1}{4\pi}(36x - 3x^2) = \frac{3x}{4\pi}(12 - x).$$

Hence $x = 12$ and $y = 6$.

Result: $\boxed{x = 12 \text{ cm}, y = 6 \text{ cm}}$, with maximum volume $\boxed{216/\pi \text{ cm}^3}$.

Problem 26

A right triangle has hypotenuse 9 cm. It is revolved about one of its legs to form a cone. Find the largest possible cone volume.

Solution 26

Let the axis leg be h and the other leg be r . Then $h^2 + r^2 = 81$ and

$$V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(81 - h^2)h.$$

Thus

$$V'(h) = \frac{\pi}{3}(81 - 3h^2) = 0,$$

so $h = 3\sqrt{3}$ and $r = 3\sqrt{6}$.

Result: The largest volume is $\boxed{54\pi\sqrt{3} \text{ cm}^3}$.

Problem 27

An isosceles triangular sheet has perimeter 40 cm. It is revolved about its base. Find the maximum possible volume of the resulting solid.

Solution 27

Let the base be b and each equal side be $\ell = (40 - b)/2$. The altitude is the radius of the solid:

$$r^2 = \ell^2 - \left(\frac{b}{2}\right)^2 = \left(20 - \frac{b}{2}\right)^2 - \frac{b^2}{4} = 400 - 20b.$$

Rotating the triangle about its base produces two cones whose combined volume is

$$V(b) = \frac{1}{3}\pi r^2 b = \frac{\pi}{3}b(400 - 20b).$$

This quadratic is maximized at $b = 10$. Then $\ell = 15$ and $r = 10\sqrt{2}$.

Result: The maximum volume is $\boxed{\frac{2000\pi}{3} \text{ cm}^3}$.

Problem 28

A circular sector has radius L and central angle θ radians. It is rolled into a right circular cone, so the sector radius becomes the cone's slant height. Find the value of θ that maximizes the cone's volume.

Solution 28

The sector arc becomes the base circumference:

$$L\theta = 2\pi r \implies r = \frac{L\theta}{2\pi}.$$

The cone height is $h = \sqrt{L^2 - r^2}$, so

$$V = \frac{\pi}{3}r^2\sqrt{L^2 - r^2}.$$

For fixed L , maximize $r^4(L^2 - r^2)$. Differentiation gives

$$2L^2 - 3r^2 = 0 \implies r = L\sqrt{\frac{2}{3}}.$$

Therefore

$$\theta = \frac{2\pi r}{L} = 2\pi\sqrt{\frac{2}{3}}.$$

Result: $\theta = 2\pi\sqrt{\frac{2}{3}} = \frac{2\pi\sqrt{6}}{3}$ radians.

Sheets, Boxes, Distance, Fencing, and Geometric Design

Problem 29

A rectangular sheet of paper has area 50 cm^2 . An advertisement is printed with margins of 1 cm at the top and bottom and $\frac{1}{2}$ cm on each side. Find the sheet dimensions that maximize the printed area.

Solution 29

Let the sheet width be x and height be y . Since $xy = 50$, $y = 50/x$. The printed dimensions are $x - 1$ and $y - 2$, so

$$A(x) = (x - 1)\left(\frac{50}{x} - 2\right) = 52 - 2x - \frac{50}{x}.$$

Then

$$A'(x) = -2 + \frac{50}{x^2} = 0 \implies x = 5,$$

and $y = 10$. Since $A''(x) = -100/x^3 < 0$, the area is maximal.

Result: The sheet should be $5 \text{ cm} \times 10 \text{ cm}$; the printed area is 32 cm^2 .

Problem 30

Squares of side x are cut from the four corners of a $12 \text{ cm} \times 12 \text{ cm}$ metal sheet. The sides are folded up to form an open box. Find x for maximum volume.

Solution 30

The box has base dimensions $(12 - 2x) \times (12 - 2x)$ and height x :

$$V(x) = x(12 - 2x)^2, \quad 0 < x < 6.$$

Then

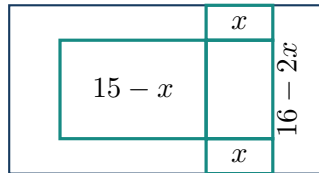
$$V'(x) = (12 - 2x)(12 - 6x).$$

The interior critical point is $x = 2$, which gives the maximum.

Result: $x = 2 \text{ cm}$. The box dimensions are $8 \times 8 \times 2 \text{ cm}$ and the maximum volume is 128 cm^3 .

Problem 31

The illustrated net of a rectangular box is cut from a $30 \text{ cm} \times 16 \text{ cm}$ rectangular sheet. If the box height is x , its other dimensions are $15 - x$ and $16 - 2x$. Find the largest possible box volume.

**Solution 31**

The volume is

$$V(x) = x(15 - x)(16 - 2x),$$

with the physical restriction $0 < x < 8$. Expanding and differentiating,

$$V'(x) = 240 - 92x + 6x^2 = 2(3x - 10)(x - 12).$$

Only $x = 10/3$ is admissible. The sign of V' changes from positive to negative there.

$$15 - x = \frac{35}{3}, \quad 16 - 2x = \frac{28}{3}.$$

Result: The dimensions are $\frac{35}{3}$ cm $\times$ $\frac{28}{3}$ cm $\times$ $\frac{10}{3}$ cm, and $V_{\max} = \frac{9800}{27}$ cm³.

Problem 32

Find the point on the curve $y = \sqrt{8x}$ that is closest to the point $(4, 2)$.

Solution 32

A point on the curve is $(x, \sqrt{8x})$, with $x \geq 0$. Minimize the squared distance

$$D^2(x) = (x - 4)^2 + (\sqrt{8x} - 2)^2.$$

Expanding,

$$D^2 = x^2 - 8x + 16 + 8x - 4\sqrt{8x} + 4 = x^2 + 20 - 8\sqrt{2x}.$$

Then

$$\frac{d}{dx}D^2 = 2x - \frac{8}{\sqrt{2x}} = 0.$$

This gives $x\sqrt{2x} = 4$, hence $x = 2$. Then $y = \sqrt{16} = 4$.

Result: The closest point is $(2, 4)$, at distance $2\sqrt{2}$.

Problem 33

A particle starts at $A = (0, 8)$ and moves down the y -axis toward the origin at 4 cm/s. At the same instant, another particle starts at $B = (10, 0)$ and moves along the positive x -axis away from the origin at 2 cm/s. When is the distance between the particles least?

Solution 33

After t seconds, the particles are

$$P(t) = (0, 8 - 4t), \quad Q(t) = (10 + 2t, 0).$$

For $0 \leq t \leq 2$, minimize

$$D^2(t) = (10 + 2t)^2 + (8 - 4t)^2.$$

Then

$$\frac{d}{dt}D^2 = 4(10 + 2t) - 8(8 - 4t) = 40t - 24.$$

Thus $t = 24/40 = 3/5$ second. The second derivative is positive.

Result: The distance is least after $\boxed{\frac{3}{5} \text{ s}}$. The minimum distance is $\boxed{\frac{28\sqrt{5}}{5} \text{ cm}}$.

Problem 34

A farmer has 1000 m of fencing for a rectangular field. The field is to be divided into three sections by two internal fences parallel to one side. Find the maximum possible area.

Solution 34

Let the side parallel to each internal fence be y , and let the other side be x . The total fencing is

$$2x + 4y = 1000 \implies x = 500 - 2y.$$

The area is

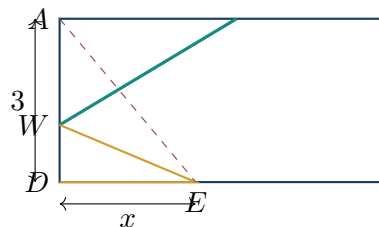
$$A(y) = xy = (500 - 2y)y = 500y - 2y^2.$$

Then $A'(y) = 500 - 4y = 0$, giving $y = 125$ and $x = 250$.

Result: The field dimensions are $\boxed{250 \text{ m} \times 125 \text{ m}}$, and the maximum area is $\boxed{31,250 \text{ m}^2}$.

Problem 35

A rectangle has height 3 cm. Its upper-left corner is folded to a point on the lower side. The crease meets the left side at W and the top side at another point. Let x be the distance from the lower-left corner D to the landing point E . Find the maximum area of the right triangle DWE .



Solution 35

Let $h = DW$. Because the upper-left corner is folded to E , the crease point W is equidistant from those two points. Hence

$$WA = WE.$$

Now $WA = 3 - h$ and $WE = \sqrt{x^2 + h^2}$, so

$$(3 - h)^2 = x^2 + h^2 \implies h = \frac{9 - x^2}{6}.$$

Therefore

$$A(x) = \frac{1}{2}xh = \frac{x(9 - x^2)}{12}, \quad 0 < x < 3.$$

Then

$$A'(x) = \frac{9 - 3x^2}{12} = 0 \implies x = \sqrt{3}.$$

Result: The maximum triangle area is $\frac{\sqrt{3}}{2} \text{ cm}^2$, attained when $DE = \sqrt{3} \text{ cm}$.

Problem 36

An isosceles trapezoid has three sides of length 20 cm. Find the length of the fourth side that maximizes the trapezoid's area.

Solution 36

Take the upper base and the two equal legs to be 20, and let the fourth side be b . Put

$$u = \frac{b - 20}{2}.$$

Then the trapezoid height is $\sqrt{20^2 - u^2}$, and $b = 20 + 2u$. Its area is

$$A(u) = \frac{20 + b}{2} \sqrt{400 - u^2} = (20 + u) \sqrt{400 - u^2}.$$

Differentiation gives

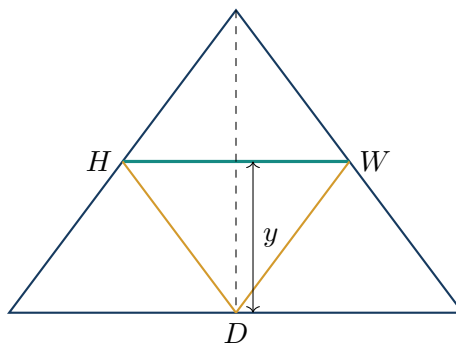
$$400 - 20u - 2u^2 = 0 \implies u = 10$$

(the other root is inadmissible). Hence $b = 40$.

Result: The fourth side should be 40 cm . The maximum area is $300\sqrt{3} \text{ cm}^2$.

Problem 37

A triangle has base 12 cm and altitude 16 cm. A segment parallel to the base cuts the other two sides. A point D lies on the original base, and the perpendicular distance from D to the parallel segment is y . Find y so that the triangle formed by D and the endpoints of the parallel segment has maximum area.



Solution 37

The small top triangle is similar to the original. If the parallel segment has length w , then

$$\frac{w}{12} = \frac{16 - y}{16}, \quad w = \frac{3}{4}(16 - y).$$

The target triangle has base w and height y :

$$A(y) = \frac{1}{2}wy = \frac{3}{8}y(16 - y).$$

Then $A'(y) = \frac{3}{8}(16 - 2y) = 0$, so $y = 8$.

Result: The required perpendicular distance is $\boxed{8 \text{ cm}}$; the maximum area is 24 cm^2 .

Problem 38

In quadrilateral $MBJD$, the sides $MB = 2 \text{ cm}$ and $MD = 1 \text{ cm}$ are fixed, and the angle $\angle BMD = x$ may vary. Triangle BJD is always right isosceles at J , so $JB = JD$. Find x that maximizes the area of the quadrilateral.

Solution 38

The quadrilateral area is the sum of triangles MBD and BJD .

$$[\triangle MBD] = \frac{1}{2}(2)(1) \sin x = \sin x.$$

By the cosine rule,

$$BD^2 = 2^2 + 1^2 - 2(2)(1) \cos x = 5 - 4 \cos x.$$

Since BJD is right isosceles with hypotenuse BD , its area is $BD^2/4$. Therefore

$$A(x) = \sin x + \frac{5 - 4 \cos x}{4} = \frac{5}{4} + \sin x - \cos x.$$

Then $A'(x) = \cos x + \sin x = 0$, giving $x = 3\pi/4$ in the geometric interval.

Result: $\boxed{x = \frac{3\pi}{4} = 135^\circ}$.

Problem 39

A triangle has two sides of lengths 5 cm and 7 cm with included angle θ . Find θ for maximum area.

Solution 39

The area is

$$A(\theta) = \frac{1}{2}(5)(7) \sin \theta = \frac{35}{2} \sin \theta.$$

This is largest when $\sin \theta = 1$.

Result: $\boxed{\theta = \frac{\pi}{2} = 90^\circ}$, and the maximum area is $35/2 \text{ cm}^2$.

Problem 40

A circle has diameter $AB = 4 \text{ cm}$. Point C moves on the circle, forming the right triangle ACB . Find $\angle ABC$ when the triangle's area is as large as possible.

Solution 40

Let $\angle ABC = \theta$. Since AB is the hypotenuse,

$$AC = 4 \sin \theta, \quad BC = 4 \cos \theta.$$

Thus

$$A(\theta) = \frac{1}{2}(4 \sin \theta)(4 \cos \theta) = 4 \sin(2\theta).$$

The maximum occurs when $2\theta = \pi/2$.

Result: $\angle ABC = \pi/4 = 45^\circ$.

Problem 41

A circular sector has perimeter 20 m. Find its radius when the sector area is as large as possible.

Solution 41

Let the radius be r and the angle be θ radians. The perimeter condition is

$$2r + r\theta = 20 \implies \theta = \frac{20}{r} - 2.$$

The area is

$$A = \frac{1}{2}r^2\theta = 10r - r^2.$$

Then $A'(r) = 10 - 2r = 0$, so $r = 5$.

Result: The radius is $\boxed{5 \text{ m}}$, the angle is 2 radians, and the maximum area is 25 m^2 .

Problem 42

An open-top box has a square base and volume 32 cm^3 . Find its dimensions so that the amount of material used is as small as possible.

Solution 42

Let the square base side be x and height be h . Since $x^2h = 32$,

$$h = \frac{32}{x^2}.$$

The material area is

$$S(x) = x^2 + 4xh = x^2 + \frac{128}{x}.$$

Then

$$S'(x) = 2x - \frac{128}{x^2} = 0 \implies x^3 = 64,$$

so $x = 4$ and $h = 2$.

Result: The box dimensions are $\boxed{4 \text{ cm} \times 4 \text{ cm} \times 2 \text{ cm}}$, with minimum material area 48 cm^2 .

Manufacturing, Cost, Revenue, Light, and Viewing Angles

Problem 43

An open-top box with a square base is made from thin wood. The total manufacturing cost is 144 dinars, and the wood costs 3 dinars per square metre. Find the largest possible box volume.

Solution 43

The available wood area is

$$\frac{144}{3} = 48 \text{ m}^2.$$

Let the square base side be x and the height be h . Because the box is open,

$$x^2 + 4xh = 48 \implies h = \frac{48 - x^2}{4x}.$$

Therefore

$$V(x) = x^2h = 12x - \frac{x^3}{4}.$$

Then $V'(x) = 12 - 3x^2/4 = 0$, so $x = 4$ and $h = 2$.

Result: The largest volume is $\boxed{32 \text{ m}^3}$, attained for a $4 \text{ m} \times 4 \text{ m}$ base and height 2 m .

Problem 44

A closed right circular cylinder is made from a metal sheet of total area $600\pi \text{ cm}^2$. Find the largest possible cylinder volume.

Solution 44

For radius r and height h ,

$$2\pi r^2 + 2\pi rh = 600\pi \implies h = \frac{300}{r} - r.$$

Hence

$$V(r) = \pi r^2 h = \pi(300r - r^3).$$

Then $V'(r) = \pi(300 - 3r^2) = 0$, giving $r = 10$ and $h = 20$.

Result: The maximum volume is $\boxed{2000\pi \text{ cm}^3}$, with radius 10 cm and height 20 cm .

Problem 45

Find the least amount of thin metal required to make a closed right circular cylindrical oil container of fixed volume 2000π litres.

Solution 45

Use decimetres, since $1 \text{ litre} = 1 \text{ dm}^3$. Then

$$\pi r^2 h = 2000\pi \implies h = \frac{2000}{r^2}.$$

The closed-cylinder area is

$$S(r) = 2\pi r^2 + 2\pi rh = 2\pi r^2 + \frac{4000\pi}{r}.$$

Then

$$S'(r) = 4\pi r - \frac{4000\pi}{r^2} = 0 \implies r^3 = 1000.$$

Thus $r = 10 \text{ dm}$ and $h = 20 \text{ dm}$.

Result: Minimum metal area = $\boxed{600\pi \text{ dm}^2}$, with radius 10 dm and height 20 dm .

Problem 46

A cylindrical water tank has capacity $3000\pi \text{ m}^3$. The base costs 4 dinars per square metre, the cylindrical side costs 2 dinars per square metre, and a hollow hemispherical cover costs 1 dinar per square metre. Find the cylinder dimensions that minimize total cost.

Solution 46

Let the cylinder radius be r and height be h . The capacity gives

$$\pi r^2 h = 3000\pi \implies h = \frac{3000}{r^2}.$$

The costs are

$$\text{base : } 4\pi r^2, \quad \text{side : } 2(2\pi r h) = 4\pi r h, \quad \text{hemisphere : } 2\pi r^2.$$

Thus

$$C(r) = 6\pi r^2 + 4\pi r \left(\frac{3000}{r^2} \right) = 6\pi r^2 + \frac{12000\pi}{r}.$$

Then

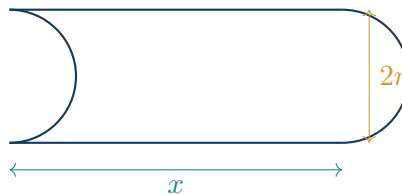
$$C'(r) = 12\pi r - \frac{12000\pi}{r^2} = 0 \implies r^3 = 1000.$$

So $r = 10$ and $h = 30$.

Result: The least-cost cylindrical part has radius 10 m and height 30 m.

Problem 47

A field is shaped like a rectangle capped by a semicircle at each end. Fencing costs 4 dinars per metre along the two straight sides and 6 dinars per metre along the curved portions. If the total fencing cost is 400 dinars, find the dimensions that maximize the field area.

**Solution 47**

Let each straight side have length x and let the semicircle radius be r . The two curved ends together form one full circle. The cost constraint is

$$4(2x) + 6(2\pi r) = 400,$$

so

$$x = 50 - \frac{3\pi r}{2}.$$

The field area is

$$A = 2rx + \pi r^2 = 2r \left(50 - \frac{3\pi r}{2} \right) + \pi r^2 = 100r - 2\pi r^2.$$

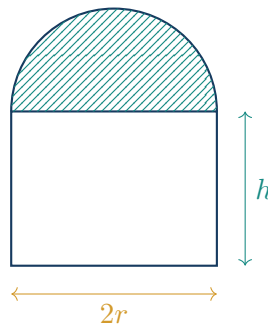
Then $A'(r) = 100 - 4\pi r = 0$, giving

$$r = \frac{25}{\pi}, \quad x = \frac{25}{2}.$$

Result: Radius = $25/\pi \text{ m}$, each straight side = $25/2 \text{ m}$, and maximum area = $1250/\pi \text{ m}^2$.

Problem 48

A window consists of a rectangle topped by a semicircle. The semicircular glass is coloured and admits only half as much light per square metre as the ordinary glass in the rectangular part. If the outside perimeter is 6 m, find the dimensions of the rectangular part that maximize the effective amount of light admitted.

**Solution 48**

Let the semicircle radius be r and rectangular height be h . The perimeter is

$$2h + 2r + \pi r = 6 \implies h = \frac{6 - (2 + \pi)r}{2}.$$

The effective light equals the clear rectangular area plus half the semicircle area:

$$L(r) = 2rh + \frac{1}{2} \left(\frac{1}{2} \pi r^2 \right) = 2rh + \frac{\pi r^2}{4}.$$

Substitution gives

$$L(r) = 6r - \left(2 + \frac{3\pi}{4} \right) r^2.$$

Then

$$L'(r) = 6 - \left(4 + \frac{3\pi}{2} \right) r = 0,$$

so

$$r = \frac{12}{8 + 3\pi}.$$

The rectangular width is $2r = 24/(8 + 3\pi)$ and

$$h = \frac{12 + 3\pi}{8 + 3\pi} = \frac{3(4 + \pi)}{8 + 3\pi}.$$

Result: Rectangle width = $\boxed{\frac{24}{8 + 3\pi} \text{ m}}$ and height = $\boxed{\frac{3(4 + \pi)}{8 + 3\pi} \text{ m}}$.

Problem 49

The monthly cost of producing x units is

$$C(x) = 3x^2 + 650,$$

and the selling price per unit is

$$p(x) = 30 - \frac{3x}{2}.$$

Find the production level that maximizes profit.

Solution 49

Revenue is

$$R(x) = xp(x) = 30x - \frac{3}{2}x^2.$$

Hence profit is

$$P(x) = R(x) - C(x) = 30x - \frac{9}{2}x^2 - 650.$$

Then

$$P'(x) = 30 - 9x = 0 \implies x = \frac{10}{3}.$$

Also $P''(x) = -9 < 0$.

Result: The continuous model is maximized at $x = 10/3$ units. If production must be an integer, 3 units gives the larger profit among the nearest choices.

Problem 50

A factory's total weekly cost, in dinars, for producing x bedroom sets is

$$C(x) = x^3 - 3x^2 - 80x + 5000.$$

Each set is sold for 2800 dinars. Find the weekly production that maximizes profit.

Solution 50

Revenue is $R(x) = 2800x$. Therefore

$$P(x) = R(x) - C(x) = -x^3 + 3x^2 + 2880x - 5000.$$

Then

$$P'(x) = -3x^2 + 6x + 2880 = -3(x - 32)(x + 30).$$

The nonnegative critical point is $x = 32$, and $P''(32) = -186 < 0$.

Result: The factory should produce 32 bedroom sets per week.

Problem 51

A factory sells x units per day at 30 dinars per unit. Its daily profit is

$$P(x) = 40x - x^2 - 100.$$

Find the production level for which the daily cost is as small as possible.

Solution 51

Revenue is $R(x) = 30x$. Since $P = R - C$,

$$C(x) = R(x) - P(x) = 30x - (40x - x^2 - 100) = x^2 - 10x + 100.$$

Then

$$C'(x) = 2x - 10 = 0 \implies x = 5,$$

and $C''(x) = 2 > 0$.

Result: The daily cost is minimized at $x = 5$ units, and the minimum cost is 75 dinars.

Problem 52

A merchant can sell 400 units of a product at a price of 1 dinar each. For every increase of 10 qirsh in the price, sales decrease by 20 units. Find the price that maximizes sales revenue.

Solution 52

Let x be the number of 0.10-dinar increases. Then

$$\text{price} = 1 + 0.1x, \quad \text{quantity} = 400 - 20x.$$

Revenue is

$$R(x) = (1 + 0.1x)(400 - 20x) = 400 + 20x - 2x^2.$$

Then $R'(x) = 20 - 4x = 0$, so $x = 5$.

Result: The best price is 1.50 dinars. At that price, 300 units are sold and revenue is 450 dinars.

Problem 53

A travel company charges each person 65 dinars when 100 people join a trip. For every additional participant above 100, the price charged to every participant decreases by 0.5 dinar. Find the number of participants that maximizes the company's revenue.

Solution 53

Let x be the number of additional participants. Then

$$R(x) = (100 + x)(65 - 0.5x) = 6500 + 15x - 0.5x^2.$$

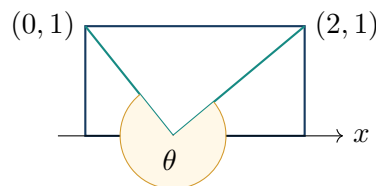
Thus

$$R'(x) = 15 - x = 0 \implies x = 15.$$

Result: The revenue is maximized with 115 participants. The price is 57.50 dinars per person and the maximum revenue is 6612.50 dinars.

Problem 54

In the rectangle with vertices $(0, 0)$, $(2, 0)$, $(2, 1)$, and $(0, 1)$, a point $J = (x, 0)$ lies on the lower side. Find x so that the angle subtended at J by the upper side is as large as possible.

**Solution 54**

Let

$$\alpha = \arctan \frac{1}{x}, \quad \beta = \arctan \frac{1}{2-x}.$$

The viewing angle is $\theta = \alpha + \beta$. Therefore

$$\theta'(x) = -\frac{1}{x^2 + 1} + \frac{1}{(2-x)^2 + 1}.$$

Setting $\theta'(x) = 0$ gives

$$x^2 + 1 = (2-x)^2 + 1,$$

so $x = 1$. Symmetry also confirms the result.

Result: $x = 1$. The optimal observation point is the midpoint of the lower side.

Problem 55

The fixed points are $A = (0, 4)$ and $B = (0, 9)$. A point $C = (x, 0)$ moves on the positive x -axis. Find the x -coordinate that maximizes $\angle ACB$.

Solution 55

The viewing angle is

$$\theta(x) = \arctan \frac{9}{x} - \arctan \frac{4}{x}.$$

Then

$$\theta'(x) = -\frac{9}{x^2 + 81} + \frac{4}{x^2 + 16}.$$

Set this equal to zero:

$$4(x^2 + 81) = 9(x^2 + 16) \implies 5x^2 = 180.$$

Hence $x = 6$.

Result: The angle is maximized when $x = 6$.

Problem 56

A billboard is 180 cm high. Its lower edge is 60 cm above an observer's eye level. How far from the wall should the observer stand to maximize the angle subtended by the billboard?

Solution 56

Relative to eye level, the bottom and top of the billboard are at heights 60 and 240 cm. If the observer stands x cm from the wall,

$$\theta(x) = \arctan \frac{240}{x} - \arctan \frac{60}{x}.$$

Then

$$\theta'(x) = -\frac{240}{x^2 + 240^2} + \frac{60}{x^2 + 60^2}.$$

Setting $\theta'(x) = 0$ yields

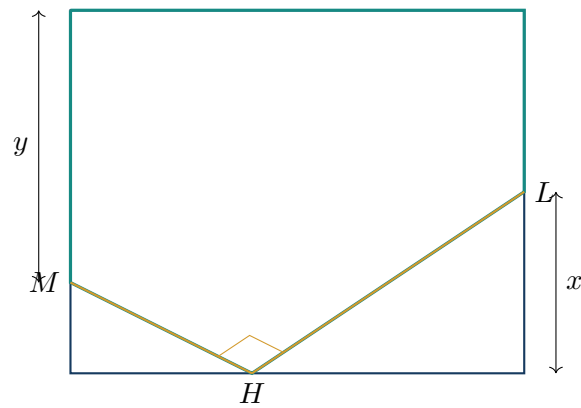
$$60(x^2 + 240^2) = 240(x^2 + 60^2),$$

which simplifies to $x^2 = 60 \cdot 240 = 14,400$.

Result: The observer should stand 120 cm from the wall.

Problem 57

A rectangle has dimensions $10 \text{ cm} \times 8 \text{ cm}$. A point H lies on the 10-cm base so that its distance from the right-hand corner is 6 cm . Points L and M lie on the right and left vertical sides, respectively, and $\angle LHM = 90^\circ$. Let $LB = x$ and let the corresponding left-side segment be y . Find x and y so that the central polygon has maximum area.

**Solution 57**

The two right triangles adjacent to H are similar because their acute angles are complementary. Their horizontal legs are 6 and 4, so

$$\frac{x}{6} = \frac{4}{y} \implies xy = 24, \quad y = \frac{24}{x}.$$

The central polygon area equals the rectangle area minus the two corner triangles:

$$A = 80 - \frac{1}{2}(6)x - \frac{1}{2}(4)y = 80 - 3x - 2y.$$

Hence

$$A(x) = 80 - 3x - \frac{48}{x}.$$

Then

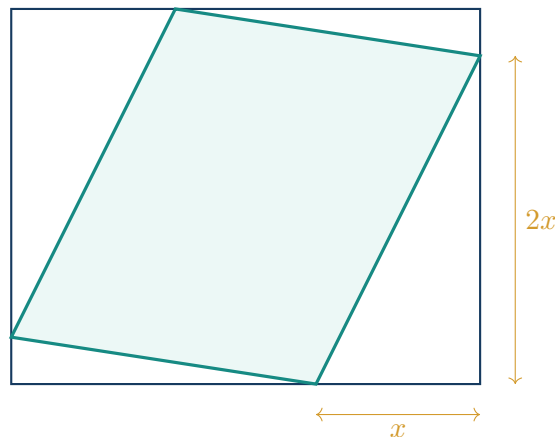
$$A'(x) = -3 + \frac{48}{x^2} = 0 \implies x = 4,$$

and $y = 6$.

Result: $x = 4$ cm, $y = 6$ cm, and the maximum central area is 56 cm².

Problem 58

A parallelogram is inscribed in a 10 cm $\times$ 8 cm rectangle, with one vertex on each side. From the lower-right corner, the horizontal distance to the bottom vertex is x , and the vertical distance to the right-side vertex is $2x$. Find x so that the parallelogram area is as large as possible.



Solution 58

Let the rectangle be $0 \leq X \leq 10$, $0 \leq Y \leq 8$. The bottom and right vertices of the parallelogram are

$$N = (10 - x, 0), \quad M = (10, 2x).$$

The parallelogram relation forces the top and left vertices to be

$$L = (x, 8), \quad W = (0, 8 - 2x).$$

Subtracting the four corner triangles from the rectangle gives

$$A(x) = 28x - 4x^2 = 4x(7 - x), \quad 0 < x < 4.$$

Then

$$A'(x) = 28 - 8x = 0 \implies x = \frac{7}{2}.$$

Result: $x = \frac{7}{2}$ cm, and the maximum parallelogram area is 49 cm^2 .

Final Answer Summary

No.	Answer	No.	Answer
1	150×150 m; $A_{\max} = 22500$ m ²	30	$x = 2$ cm; $V_{\max} = 128$ cm ³
2	20×20 m; $P_{\min} = 80$ m	31	$\frac{35}{3} \times \frac{28}{3} \times \frac{10}{3}$ cm
3	(a) 10, 10; (b) $\frac{20}{3}, \frac{40}{3}$	32	Closest point (2, 4)
4	$x = 3$	33	$t = \frac{3}{5}$ s
5	$a_{\min} = 48$ m/s ² at $t = 3$ s	34	250×125 m
6	4×8 ; area 32	35	$A_{\max} = \frac{\sqrt{3}}{2}$ cm ²
7	4×8 ; area 32	36	Fourth side 40 cm
8	4×24 ; area 96	37	Distance 8 cm
9	6×8 ; area 48	38	$x = 135^\circ$
10	Area 54	39	$\theta = 90^\circ$
11	Bases 6, 2; height 8	40	45°
12	$5\sqrt{2} \times 5/\sqrt{2}$ cm	41	$r = 5$ m
13	4×5 cm	42	$4 \times 4 \times 2$ cm
14	$x/(2a) + y/(2b) = 1$	43	$4 \times 4 \times 2$ m
15	$3x + 2y = 12$	44	$r = 10$, $h = 20$ cm
16	$r = 10/3$, $h = 3$ cm	45	$r = 10$, $h = 20$ dm
17	Cylinder volume = $\frac{4}{9}$ cone volume	46	$r = 10$, $h = 30$ m
18	$r = 28/3$, $h = 56/9$ cm	47	$r = 25/\pi$, $x = 25/2$ m
19	Cone radius $6\sqrt{2}$ cm	48	Width $24/(8 + 3\pi)$ m
20	Cylinder height 6 cm	49	$x = 10/3$
21	Cylinder radius 10 cm	50	$x = 32$ sets
22	Cube side $20/\sqrt{3}$ cm	51	$x = 5$ units
23	$V_{\max} = 256\pi/27$ cm ³	52	Price 1.50 dinars
24	$V_{\max} = 2000\pi/(3\sqrt{3})$ cm ³	53	115 participants
25	$x = 12$, $y = 6$ cm	54	$x = 1$
26	$V_{\max} = 54\pi\sqrt{3}$ cm ³	55	$x = 6$
27	$V_{\max} = 2000\pi/3$ cm ³	56	Distance 120 cm
28	$\theta = 2\pi\sqrt{2/3}$	57	$x = 4$, $y = 6$ cm
29	Sheet 5×10 cm	58	$x = 7/2$ cm; area 49 cm ²

End of Complete English Edition

All 58 questions include a mathematical model, calculus optimization, and final answer.